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T11 – 05

Assignment No. 9

Queries:

- Sort by year (ascending) and find the oldest car
- Make of the car that is black and built in 1977.
- Count of Ford cars
- Count of cars costing less than Rs. 100000
- Update color to red where brand is "jaguar"

LO5: Create REST Web services using MongoDB

THEORY:

In MongoDB's command line interface (CLI), users interact directly with the database using simple commands. Starting with the ``mongo`` command, users connect to MongoDB and select a specific database using ``use <database_name>``. Querying data involves the straightforward ``db.<collection_name>.find()`` command, while updating documents is achieved through ``db.<collection_name>.update()``. For more complex operations, such as aggregation, users employ ``db.<collection_name>.aggregate()``. Miscellaneous tasks, like deleting documents or counting entries, are handled with appropriate commands. Once tasks are completed, users exit the MongoDB CLI by typing ``exit``. This direct approach allows for efficient management and manipulation of data without the need for additional tools or interfaces.

SCREENSHOTS:

- Sort by year (ascending) and find the oldest car

myCompiler

English Recent Login

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MongoDB ⓘ

▶ Run

1 db.Car.insertMany([{ brand: "ford", make: "mustang", color: "blue", year: 1980, price: 150000 }, { brand: "jaguar", make: "xj" }

2

Output

```
mycompiler_mongodb> {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('67feb8ae58ce37e5476b128c'),
    '1': ObjectId('67feb8ae58ce37e5476b128d'),
    '2': ObjectId('67feb8ae58ce37e5476b128e'),
    '3': ObjectId('67feb8ae58ce37e5476b128f'),
    '4': ObjectId('67feb8ae58ce37e5476b1290'),
    '5': ObjectId('67feb8ae58ce37e5476b1291'),
    '6': ObjectId('67feb8ae58ce37e5476b1292'),
    '7': ObjectId('67feb8ae58ce37e5476b1293'),
    '8': ObjectId('67feb8ae58ce37e5476b1294'),
    '9': ObjectId('67feb8ae58ce37e5476b1295')
  }
}
mycompiler_mongodb>
```

Enter a title...

MongoDB ⓘ

▶ Run

1 db.Car.insertMany([{ brand: "ford", make: "mustang", color: "blue", year: 1980, price: 150000 }, { brand: "jaguar", make: "xj", color: "black", year: 1977, price: 95000 }])

2 db.Car.find().sort({ year: 1 })

3

Output

```
mycompiler_mongodb> [
  {
    _id: ObjectId('67feb8d39a696510ba6b128d'),
    brand: 'jaguar',
    make: 'xj',
    color: 'black',
    year: 1977,
    price: 95000
  },
  {
    _id: ObjectId('67feb8d39a696510ba6b128c'),
    brand: 'ford',
    make: 'mustang',
    color: 'blue',
    year: 1980,
    price: 150000
  },
  {
    _id: ObjectId('67feb8d39a696510ba6b128e'),
    brand: 'ford',
    make: 'mustang',
    color: 'blue',
    year: 1980,
    price: 150000
  },
  {
    _id: ObjectId('67feb8d39a696510ba6b128f'),
    brand: 'jaguar',
    make: 'xj',
    color: 'black',
    year: 1977,
    price: 95000
  },
  {
    _id: ObjectId('67feb8d39a696510ba6b1290'),
    brand: 'ford',
    make: 'mustang',
    color: 'blue',
    year: 1980,
    price: 150000
  },
  {
    _id: ObjectId('67feb8d39a696510ba6b1291'),
    brand: 'ford',
    make: 'mustang',
    color: 'blue',
    year: 1980,
    price: 150000
  },
  {
    _id: ObjectId('67feb8d39a696510ba6b1292'),
    brand: 'ford',
    make: 'mustang',
    color: 'blue',
    year: 1980,
    price: 150000
  },
  {
    _id: ObjectId('67feb8d39a696510ba6b1293'),
    brand: 'ford',
    make: 'mustang',
    color: 'blue',
    year: 1980,
    price: 150000
  },
  {
    _id: ObjectId('67feb8d39a696510ba6b1294'),
    brand: 'ford',
    make: 'mustang',
    color: 'blue',
    year: 1980,
    price: 150000
  },
  {
    _id: ObjectId('67feb8d39a696510ba6b1295'),
    brand: 'ford',
    make: 'mustang',
    color: 'blue',
    year: 1980,
    price: 150000
  }
]
```

```
{
  _id: ObjectId('67feb8d39a696510ba6b1290'),
  brand: 'toyota',
  make: 'corolla',
  color: 'black',
  year: 2005,
  price: 90000
},
{
  _id: ObjectId('67feb8d39a696510ba6b1291'),
  brand: 'maruti',
  make: 'alto',
  color: 'white',
  year: 2008,
  price: 60000
},
{
  _id: ObjectId('67feb8d39a696510ba6b128f'),
  brand: 'honda',
  make: 'civic',
  color: 'red',
  year: 2010,
  price: 85000
},
{
  _id: ObjectId('67feb8d39a696510ba6b128e'),
  brand: 'ford',
  make: 'figo',
  color: 'white',
  year: 2012,
  price: 50000
},
{
  _id: ObjectId('67feb8d39a696510ba6b1295'),
  brand: 'audi',
  make: 'a8',
  color: 'black',
  year: 2015,
  price: 120000
}
```

```
{
  _id: ObjectId('67feb8d39a696510ba6b1295'),
  brand: 'audi',
  make: 'a4',
  color: 'blue',
  year: 2015,
  price: 400000
},
{
  _id: ObjectId('67feb8d39a696510ba6b1292'),
  brand: 'ford',
  make: 'ecosport',
  color: 'gray',
  year: 2016,
  price: 300000
},
{
  _id: ObjectId('67feb8d39a696510ba6b1294'),
  brand: 'bmw',
```

```

{
  _id: ObjectId('67feba04c23923ffee6b1294'),
  brand: 'bmw',
  make: 'x5',
  color: 'black',
  year: 2018,
  price: 550000
},
{
  _id: ObjectId('67feba04c23923ffee6b1293'),
  brand: 'jaguar',
  make: 'f-type',
  color: 'green',
  year: 2020,
  price: 600000
}
]
mycompiler_mongodb>
```

- Make of the car that is black and built in 1977.

MongoDB

```
1 db.Car.insertMany([ { brand: "ford", make: "mustang", color: "blue", year:
2 db.Car.find( { color: "black", year: 1977 }, { make: 1, _id: 0 } )
3
4
```

Output

```
mycompiler_mongodb> {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('67feba5cda064294b86b128c'),
    '1': ObjectId('67feba5cda064294b86b128d'),
    '2': ObjectId('67feba5cda064294b86b128e'),
    '3': ObjectId('67feba5cda064294b86b128f'),
    '4': ObjectId('67feba5cda064294b86b1290'),
    '5': ObjectId('67feba5cda064294b86b1291'),
    '6': ObjectId('67feba5cda064294b86b1292'),
    '7': ObjectId('67feba5cda064294b86b1293'),
    '8': ObjectId('67feba5cda064294b86b1294'),
    '9': ObjectId('67feba5cda064294b86b1295')
  }
}
mycompiler_mongodb> [ { make: 'xj' } ]
mycompiler_mongodb>
```

Count of Ford cars

MongoDB

```
1 db.Car.insertMany([ { brand: "ford", make: "mustang", color: "blue", year:
2 db.Car.countDocuments( { brand: "ford" } )
3
4
5
```

Output

```
mycompiler_mongodb> {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('67febad039d9154ba36b128c'),
    '1': ObjectId('67febad039d9154ba36b128d'),
    '2': ObjectId('67febad039d9154ba36b128e'),
    '3': ObjectId('67febad039d9154ba36b128f'),
    '4': ObjectId('67febad039d9154ba36b1290'),
    '5': ObjectId('67febad039d9154ba36b1291'),
    '6': ObjectId('67febad039d9154ba36b1292'),
    '7': ObjectId('67febad039d9154ba36b1293'),
    '8': ObjectId('67febad039d9154ba36b1294'),
    '9': ObjectId('67febad039d9154ba36b1295')
  }
}
mycompiler_mongodb> 3
mycompiler_mongodb>
```

Count of cars costing less than Rs. 100000

MongoDB



```
1 db.Car.insertMany([ { brand: "ford", make: "mustang", color: "blue", year: 2018 },
2 { brand: "ford", make: "mustang", color: "blue", year: 2019 },
3 { brand: "ford", make: "mustang", color: "blue", year: 2020 },
4 { brand: "ford", make: "mustang", color: "blue", year: 2021 },
5 { brand: "ford", make: "mustang", color: "blue", year: 2022 },
6 { brand: "ford", make: "mustang", color: "blue", year: 2023 } ])
```

Output

```
mycompiler_mongodb> {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('67feba77abd588ea926b128c'),
    '1': ObjectId('67feba77abd588ea926b128d'),
    '2': ObjectId('67feba77abd588ea926b128e'),
    '3': ObjectId('67feba77abd588ea926b128f'),
    '4': ObjectId('67feba77abd588ea926b1290'),
    '5': ObjectId('67feba77abd588ea926b1291'),
    '6': ObjectId('67feba77abd588ea926b1292'),
    '7': ObjectId('67feba77abd588ea926b1293'),
    '8': ObjectId('67feba77abd588ea926b1294'),
    '9': ObjectId('67feba77abd588ea926b1295')
  }
}
mycompiler_mongodb> 5
mycompiler_mongodb>
```

- Update colour to red where brand is "jaguar"

```
1 db.Car.insertMany([ { brand: "ford", make: "mustang", color: "blue", year: 2018 },
2 { brand: "ford", make: "mustang", color: "blue", year: 2019 },
3 { brand: "jaguar", make: "xj", color: "blue", year: 2018 },
4 { brand: "jaguar", make: "xj", color: "blue", year: 2019 },
5 { brand: "jaguar", make: "xj", color: "blue", year: 2020 },
6 { brand: "jaguar", make: "xj", color: "blue", year: 2021 },
7 { brand: "jaguar", make: "xj", color: "blue", year: 2022 },
8 { brand: "jaguar", make: "xj", color: "blue", year: 2023 },
9 { brand: "jaguar", make: "xj", color: "blue", year: 2024 },
10 { brand: "jaguar", make: "xj", color: "blue", year: 2025 } ])
```

Output

```
mycompiler_mongodb> {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('67febb2855d23411e16b128c'),
    '1': ObjectId('67febb2855d23411e16b128d'),
    '2': ObjectId('67febb2855d23411e16b128e'),
    '3': ObjectId('67febb2855d23411e16b128f'),
    '4': ObjectId('67febb2855d23411e16b1290'),
    '5': ObjectId('67febb2855d23411e16b1291'),
    '6': ObjectId('67febb2855d23411e16b1292'),
    '7': ObjectId('67febb2855d23411e16b1293'),
    '8': ObjectId('67febb2855d23411e16b1294'),
    '9': ObjectId('67febb2855d23411e16b1295')
  }
}
mycompiler_mongodb> ... .. {
  acknowledged: true,
  insertedId: null,
  matchedCount: 2,
  modifiedCount: 2,
  upsertedCount: 0
}
```

CONCLUSION:

In this assignment, we successfully created a car database containing key attributes such as Brand, Make, Colour, Year, and Price. The collection was populated with at least ten documents, representing a variety of car models and manufacturers. This exercise enhanced our understanding of designing a structured and meaningful database schema, inserting multiple records efficiently, and preparing the dataset for potential use in vehicle inventory systems or automotive applications.